SR UNIVERSITY

## School of Computer Science and Artificial Intelligence

# Lab Assignment 14.5

| | |
|---|---|
| **Program** | **: B. Tech (CSE)** |
| **Specialization** | **: -** |
| **Course Title** | **: AI Assisted Coding** |
| **Course Code** | **: 23CS002PC304** |
| **Semester** | **:II** |
| **Academic Session** | **: 2025-2026** |
| **Name of Student** | **: R.Sai Kowshik** |
| **Enrollment No.** | **: 2403A51L23** |
| **Batch No :** | **: 51** |
| **Date** | **: 13/03/26** |

## Task 1 – Image Gallery Website

### Prompt:

Create an image gallery web page.

### Requirements:

• Display images in a grid layout.

• Use AI to:

- Optimize layout using CSS Grid.
- Add hover zoom or overlay effects.
- Ensure mobile responsiveness.

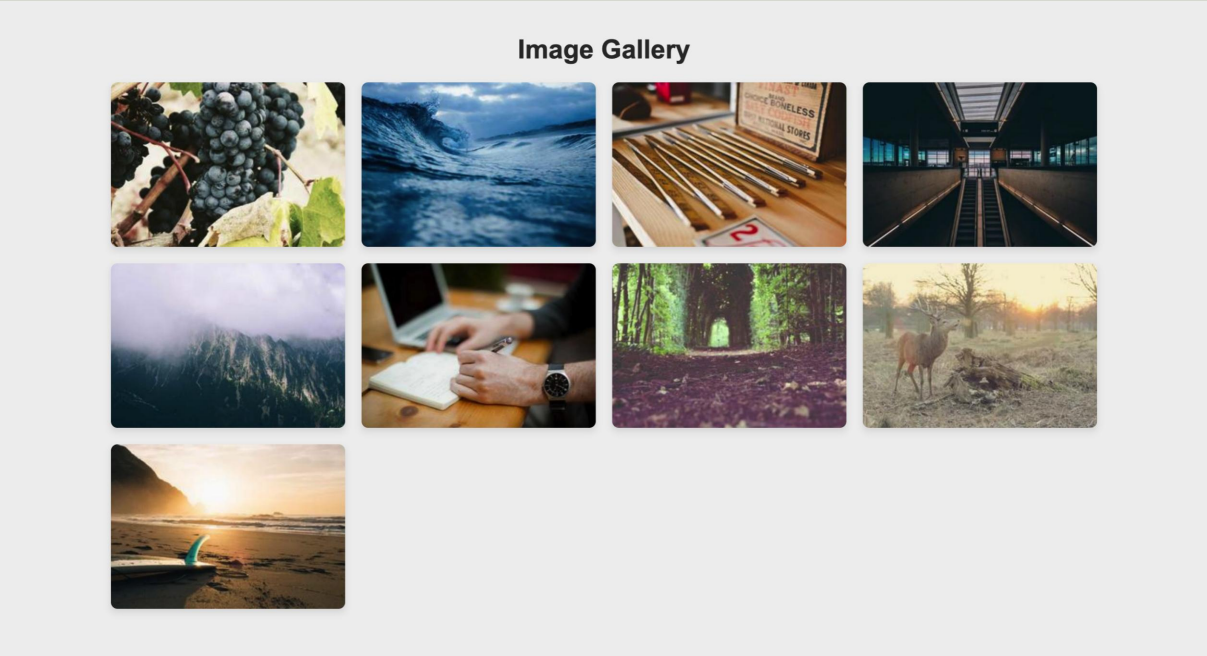
### Code:

index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Image Gallery</title>
    <link rel="stylesheet" href="styles.css">
</head>
<body>
    <h1>Image Gallery</h1>
    <div class="gallery">
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=1" alt="Random Image 1">
            <div class="overlay">
                <p>Image 1</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=2" alt="Random Image 2">
            <div class="overlay">
                <p>Image 2</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=3" alt="Random Image 3">
            <div class="overlay">
                <p>Image 3</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=4" alt="Random Image 4">
            <div class="overlay">
                <p>Image 4</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=5" alt="Random Image 5">
            <div class="overlay">
                <p>Image 5</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=6" alt="Random Image 6">
            <div class="overlay">
                <p>Image 6</p>
```

```html
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=7" alt="Random Image 7">
            <div class="overlay">
                <p>Image 7</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=8" alt="Random Image 8">
            <div class="overlay">
                <p>Image 8</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=9" alt="Random Image 9">
            <div class="overlay">
                <p>Image 9</p>
            </div>
        </div>
    </div>
</body>
</html>
```

## Output:



SR UNIVERSITY

# Task 2 – Profile Card Generator

**Prompt:** Create a profile card web page.

## Requirements:

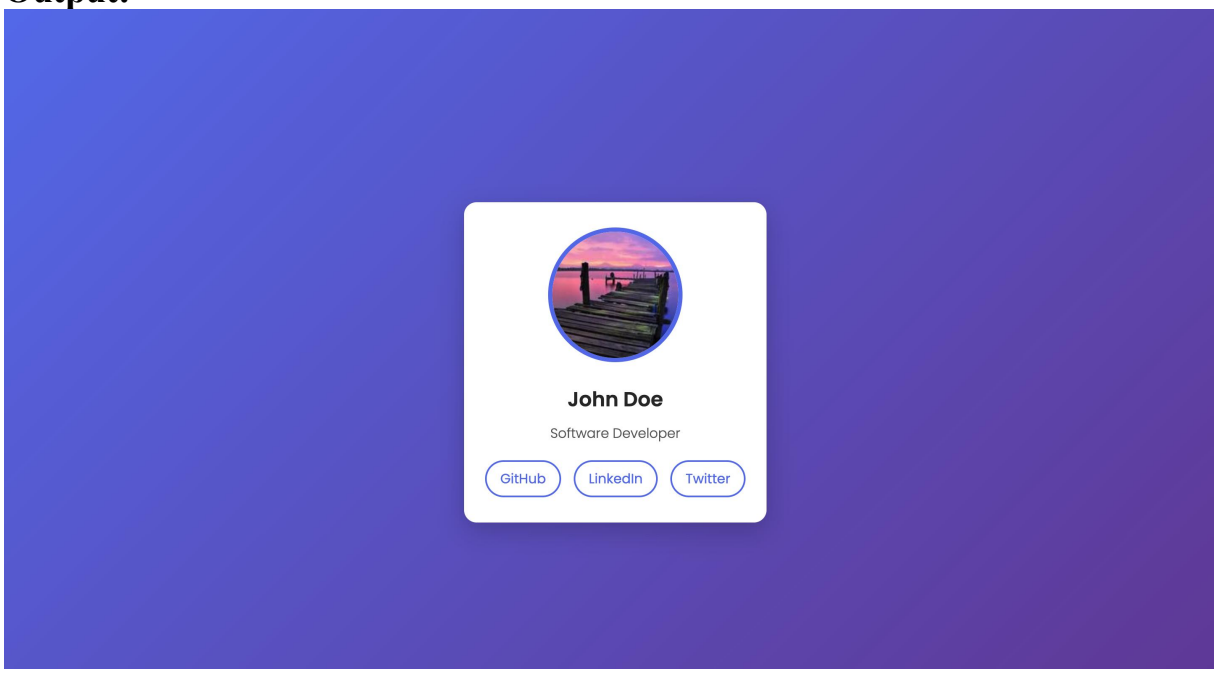
• Display user image, name, role, and social media links.
• Use AI to:
- Suggest modern UI colors and fonts.
- Design the card using Flexbox.
- Add hover animation and shadow effects.

## Code:

profile.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Profile Card</title>
    <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=
    <link rel="stylesheet" href="profile.css">
</head>
<body>
    <div class="profile-card">
        <div class="profile-image">
            <img src="https://picsum.photos/150/150?random=10" alt="Profile Picture">
        </div>
        <div class="profile-info">
            <h1 class="name">Kowshik</h1>
            <p class="role">Software Developer</p>
            <div class="social-links">
                <a href="#" class="social-link" data-platform="github">GitHub</a>
                <a href="#" class="social-link" data-platform="linkedin">LinkedIn</a>
                <a href="#" class="social-link" data-platform="twitter">Twitter</a>
            </div>
        </div>
    </div>
</body>
</html>
```

**Output:**




# Task 3 -Develop a dynamic shopping cart system.

**Prompt:** Create a shopping cart system.

## Requirements:

- Add, remove, and update product quantities.
- Display real-time total price.
- Use AI to:
  - Manage cart state using JavaScript objects.
  - Persist cart data in localStorage.

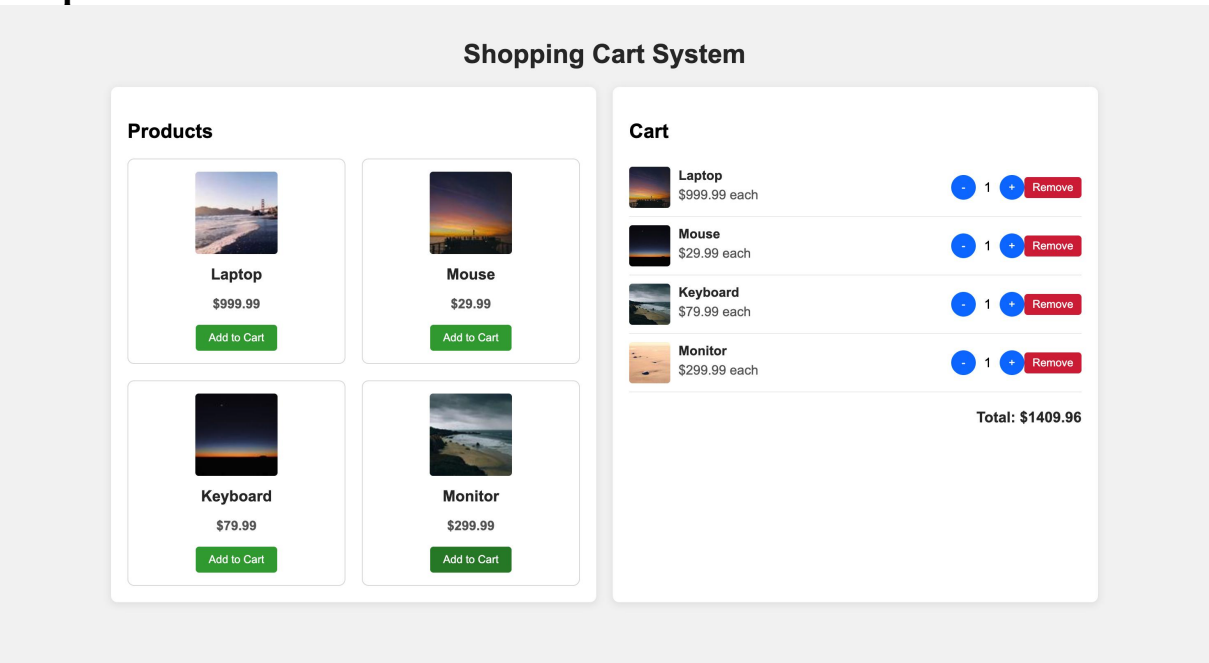
## Code:

cart.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Shopping Cart</title>
    <link rel="stylesheet" href="cart.css">
</head>
<body>
    <h1>Shopping Cart System</h1>
    <div class="container">
        <div class="products">
            <h2>Products</h2>
            <div class="product-list">
                <div class="product" data-id="1">
                    <img src="https://picsum.photos/100/100?random=20" alt="Product 1">
                    <h3>Laptop</h3>
                    <p>$999.99</p>
                    <button class="add-to-cart">Add to Cart</button>
                </div>
                <div class="product" data-id="2">
                    <img src="https://picsum.photos/100/100?random=21" alt="Product 2">
                    <h3>Mouse</h3>
                    <p>$29.99</p>
                    <button class="add-to-cart">Add to Cart</button>
                </div>
                <div class="product" data-id="3">
                    <img src="https://picsum.photos/100/100?random=22" alt="Product 3">
                    <h3>Keyboard</h3>
                    <p>$79.99</p>
                    <button class="add-to-cart">Add to Cart</button>
                </div>
                <div class="product" data-id="4">
                    <img src="https://picsum.photos/100/100?random=23" alt="Product 4">
                    <h3>Monitor</h3>
                    <p>$299.99</p>
                    <button class="add-to-cart">Add to Cart</button>
                </div>
            </div>
        </div>
        <div class="cart">
            <h2>Cart</h2>
            <div class="cart-items"></div>
            <div class="cart-total">
```

```html
                <strong>Total: $<span id="total">0.00</span></strong>
            </div>
        </div>
    </div>
    <script src="cart.js"></script>
</body>
</html>
```

**Output:**




# Task 4 – Role-Based Access Control Application

**Prompt:** Create a role-based web application.

**Requirements:**

• Admin, Editor, User roles.
• Conditional UI rendering.
• Use AI to:
  - Design access control logic.
  - Maintain secure state transitions.
  - Create scalable UI components.

SR UNIVERSITY

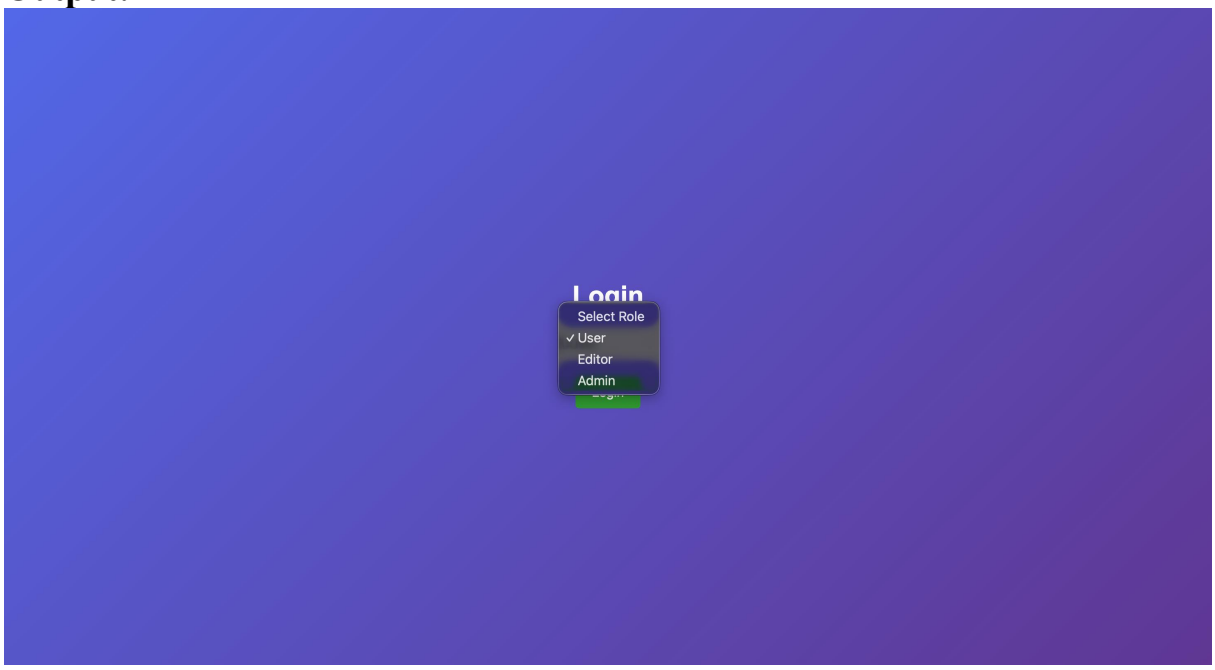
**Code:**

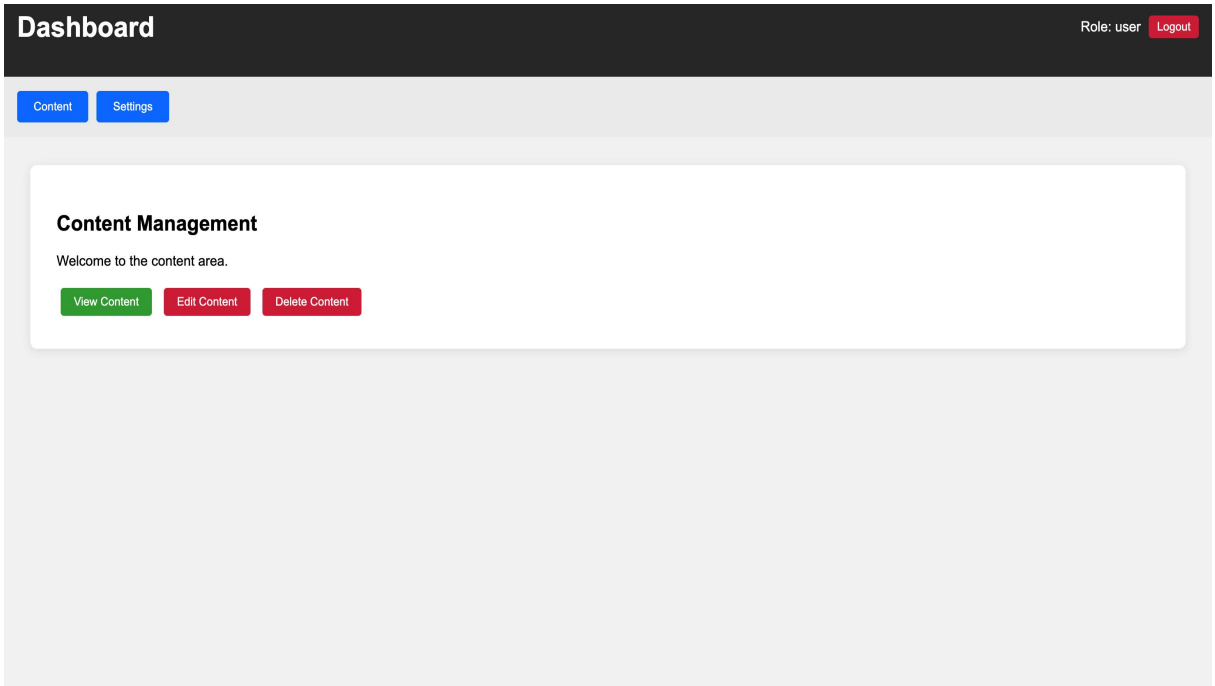
rbac.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Role-Based Access Control</title>
    <link rel="stylesheet" href="rbac.css">
</head>
<body>
    <div id="login" class="login-container">
        <h1>Login</h1>
        <select id="role-select">
            <option value="">Select Role</option>
            <option value="user">User</option>
            <option value="editor">Editor</option>
            <option value="admin">Admin</option>
        </select>
        <button id="login-btn">Login</button>
    </div>

    <div id="dashboard" class="dashboard" style="display: none;">
        <header class="header">
            <h1>Dashboard</h1>
            <div class="user-info">
                <span id="current-role">Role: </span>
                <button id="logout-btn">Logout</button>
            </div>
        </header>

        <nav class="nav">
            <button class="nav-btn" data-section="content">Content</button>
            <button class="nav-btn" data-section="users" id="users-nav">Users</button>
            <button class="nav-btn" data-section="settings" id="settings-nav">Settings</b
        </nav>

        <main class="main">
            <section id="content-section" class="section">
                <h2>Content Management</h2>
                <p>Welcome to the content area.</p>
                <button id="view-content-btn" class="action-btn">View Content</button>
                <button id="edit-content-btn" class="action-btn">Edit Content</button>
                <button id="delete-content-btn" class="action-btn">Delete Content</button
            </section>

```

```html
            <section id="users-section" class="section" style="display: none;">
                <h2>User Management</h2>
                <p>Manage users here.</p>
                <button id="view-users-btn" class="action-btn">View Users</button>
                <button id="add-user-btn" class="action-btn">Add User</button>
                <button id="delete-user-btn" class="action-btn">Delete User</button>
            </section>

            <section id="settings-section" class="section" style="display: none;">
                <h2>Settings</h2>
                <p>Application settings.</p>
                <button id="view-settings-btn" class="action-btn">View Settings</butto
                <button id="edit-settings-btn" class="action-btn">Edit Settings</butto
            </section>
        </main>
    </div>

    <script src="rbac.js"></script>
</body>
</html>
```

# Output:


Login
Select Role
User
Editor
Admin


Dashboard
Role: user
Logout
Content
Settings
Content Management
Welcome to the content area.
View Content
Edit Content
Delete Content

SR UNIVERSITY

# Task 5 – Multi-Language Website

**Prompt:**Design a multilingual website.

## Requirements:

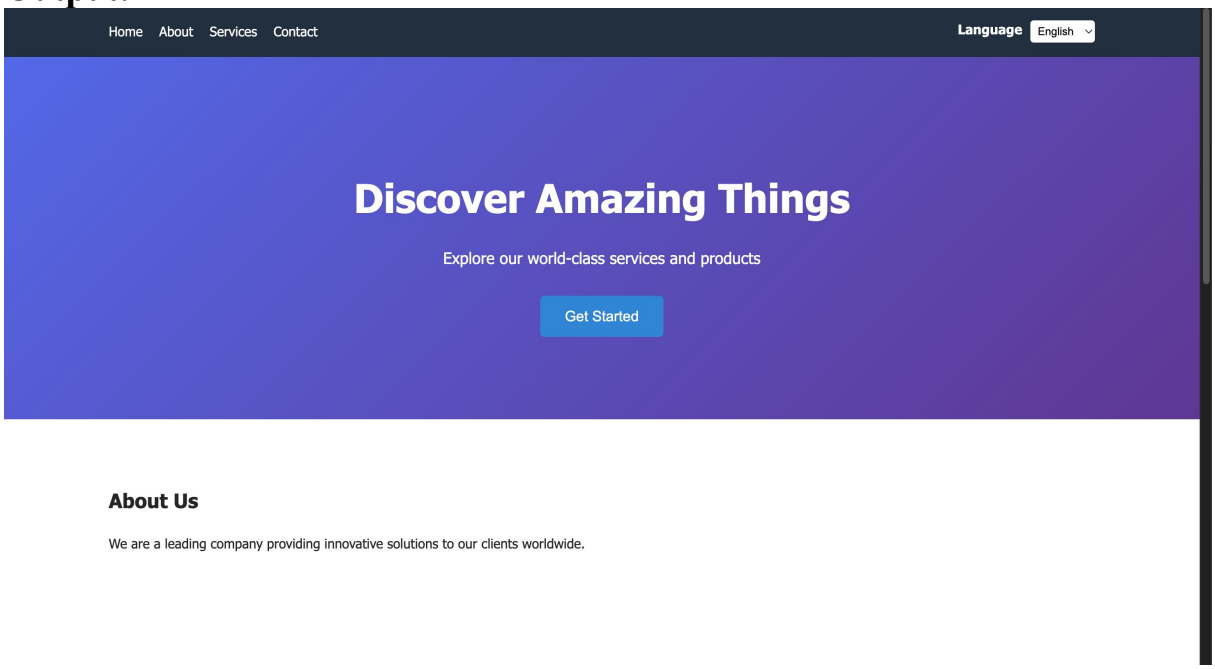
- Switch languages dynamically.
- Persist language preference.
- Use AI to:
  - Structure translation files.
  - Handle RTL/LTR layouts.
  - Improve accessibility.

## Code:

```html
<!DOCTYPE html>
<html lang="en" dir="ltr">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title data-i18n="title">Welcome to Our Website</title>
    <link rel="stylesheet" href="multi.css">
</head>
<body>
    <header>
        <nav>
            <div class="nav-container">
                <ul>
                    <li><a href="#" data-i18n="nav.home">Home</a></li>
                    <li><a href="#" data-i18n="nav.about">About</a></li>
                    <li><a href="#" data-i18n="nav.services">Services</a></li>
                    <li><a href="#" data-i18n="nav.contact">Contact</a></li>
                </ul>
                <div class="lang-switcher">
                    <label for="lang-select" data-i18n="lang">Language:</label>
                    <select id="lang-select" aria-label="Select language">
                        <option value="en">English</option>
                        <option value="es">Español</option>
                        <option value="ar">العربية</option>
                    </select>
                </div>
            </div>
        </nav>
    </header>

    <main>
        <section class="hero">
            <div class="hero-container">
                <h1 data-i18n="hero.heading">Discover Amazing Things</h1>
                <p data-i18n="hero.subheading">Explore our world-class services and produ
                <button data-i18n="hero.cta">Get Started</button>
            </div>
        </section>

        <section class="about">
            <div class="container">
                <h2 data-i18n="about.title">About Us</h2>
                <p data-i18n="about.content">We are a leading company providing innovativ
            </div>
```

```html

        <section class="services">
            <div class="container">
                <h2 data-i18n="services.title">Our Services</h2>
                <div class="services-grid">
                    <div class="service">
                        <h3 data-i18n="services.web">Web Development</h3>
                        <p>Custom web solutions tailored to your needs.</p>
                    </div>
                    <div class="service">
                        <h3 data-i18n="services.design">Graphic Design</h3>
                        <p>Creative designs that capture your brand essence.</p>
                    </div>
                    <div class="service">
                        <h3 data-i18n="services.consulting">Consulting</h3>
                        <p>Expert advice to drive your business forward.</p>
                    </div>
                </div>
            </div>
        </section>

        <section class="contact">
            <div class="container">
                <h2 data-i18n="contact.title">Contact Us</h2>
                <form>
                    <div class="form-group">
                        <label for="name" data-i18n="contact.name">Name</label>
                        <input type="text" id="name" name="name" required aria-required="
                    </div>
                    <div class="form-group">
                        <label for="email" data-i18n="contact.email">Email</label>
                        <input type="email" id="email" name="email" required aria-require
                    </div>
                    <div class="form-group">
                        <label for="message" data-i18n="contact.message">Message</label>
                        <textarea id="message" name="message" required aria-required="tru
                    </div>
                    <button type="submit" data-i18n="contact.send">Send</button>
                </form>
            </div>
        </section>
    </main>
```

**Output:**




# Task 6 – Online Examination System

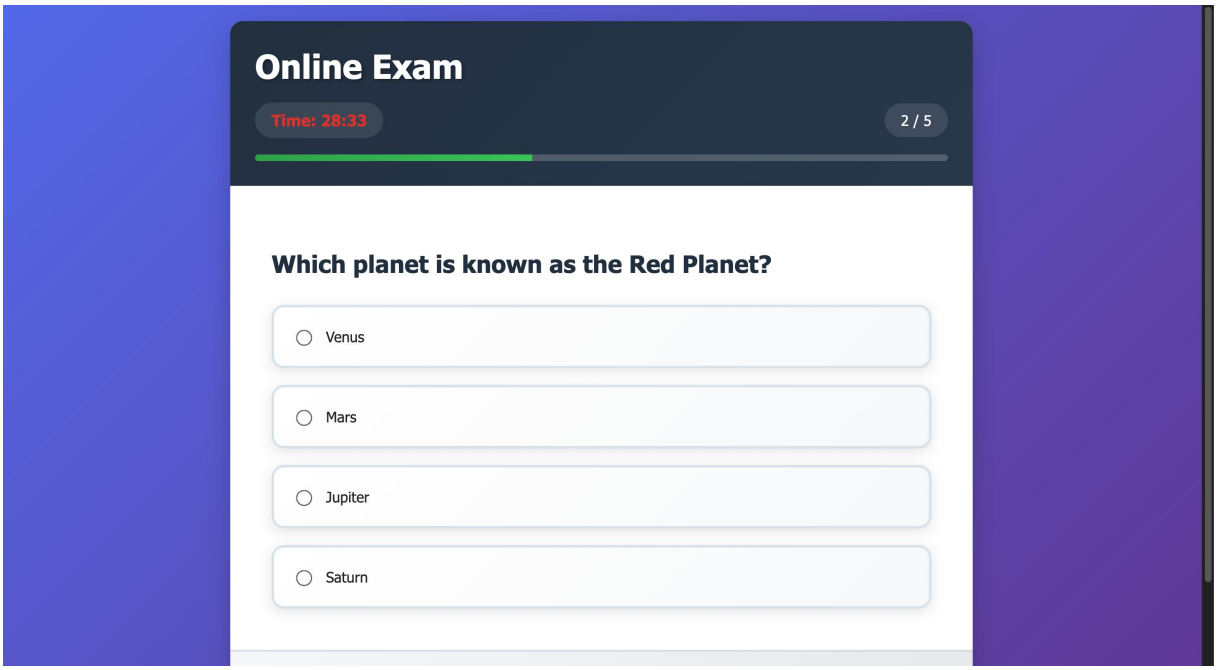
**Prompt:**Create an online exam interface.

**Requirements:**

• Timer, questions, and submission.
• Prevent multiple submissions.
• Use AI to:
  - Manage exam state.
  - Improve exam UX.
  - Ensure accessibility.

SR UNIVERSITY

## Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Online Examination System</title>
    <link rel="stylesheet" href="exam.css">
</head>
<body>
    <div id="exam-container">
        <header>
            <h1>Online Exam</h1>
            <div class="exam-info">
                <div class="timer" id="timer">Time: 30:00</div>
                <div class="progress">
                    <span id="current-question">1</span> / <span id="total-questions">5
                </div>
            </div>
        </header>

        <main>
            <div class="question-container" id="question-container">
                <div class="question" id="question">
                    <h2 id="question-text">Question text here</h2>
                    <div class="options" id="options">
                        <!-- Options will be dynamically added -->
                    </div>
                </div>
            </div>

            <div class="navigation">
                <button id="prev-btn" disabled>Previous</button>
                <button id="next-btn">Next</button>
                <button id="submit-btn" style="display: none;">Submit Exam</button>
            </div>
        </main>
    </div>

    <div id="results" style="display: none;">
        <h2>Exam Completed</h2>
        <p>Your score: <span id="score"></span></p>
        <button id="retake-btn">Retake Exam</button>
    </div>

    <script src="exam.js"></script>
```

## Output: